

Tennis Secrets: Momentum and Match Swing Analysis

Summary

The Wimbledon Open is a long-established and highly regarded tennis tournament. In the men's final of Wimbledon 2023, 20-year-old Spanish star Carlos Alcaraz surprisingly defeated 36-year-old veteran Novak Djokovic. In this paper, through the establishment of the momentum model, **DTMM model**, and **Random Forest model**, a series of conclusions in favor of predicting the direction of the match are obtained. In turn, it can help players better adjust their status.

Firstly, in order to analyze the flow of the game when capturing scores and quantitatively describe the performance of a player over some time, we analyze the data of tennis matches, get the factors affecting the **momentum**, use the entropy weighting method to find out their weights and derive a quantitative description of the momentum, and the weights of the influencing factors are shown in Fig.3. Subsequently, we combined the momentum with the other influencing factors. We constructed a DTMM (decision tree momentum model) model with other influencing factors and carried out the prediction of game score data. We found that momentum has a decisive role in prediction accuracy. Finally, a sensitivity test was conducted, and it was found that the DTMM model is sensitive to momentum, and the tree depth should be chosen properly.

Secondly, we summed the player scores over a window of time and used it to represent the players' overall performance over time. Based on this data, we conducted a **run test** and found that momentum is not random, while player performance is random, as shown in Table 3. We also applied the control variable method and ran the DTMM model with and without momentum. In the absence of the momentum effect, there was a significant dip in the accuracy of the predictions, reflecting that momentum plays a crucial role in player performance.

Thirdly, to accurately analyze when on-court fluctuations shift from favoring one player to another, we optimized the DTMM model using a random forest model. Using this model, we defined the match fluctuation **GON** (see Equation (4)) and predicted it using Novak Djokovic as an example, and found that his performance was excellent in most of the matches and dipped in this 1701, which is consistent with the facts, as shown in Fig.11 and 12. Subsequently, we utilized the SHAP algorithm to analyze the individual influencing factors. Based on this, we gave Novak Djokovic a technical and tactical analysis and training recommendations.

Lastly, to test the performance of the DTMM model in predicting fluctuations in other tournaments, we found and trained on data from the 2023 US Open.

Keywords: Momentum, DTMM, Random Forest Model, GON, tennis

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1 Introduction

1.1 Background

Tennis, a sport with a long history, is a tense game of endurance, speed, and reflexes. Players strive for victory and physical fitness by hitting the ball to the opponent's court while avoiding a triumphant return.

The Wimbledon Championships is the longest-running and most prestigious open event in the sport, and Novak Djokovic has been a Grand Slam player since 2013, dominating the men's singles track at the Wimbledon Championships for more than a decade. However, tennis is dynamic, and there is always enough young talent to inject fresh blood into it. In 2023, at 20, Spain's Carlos Alcaraz became a rising star when he beat 36-year-old Novak Djokovic 3-2 in the men's singles championship.

Many factors influence the direction of a match, and momentum, as a subjective force that is difficult to measure, plays a vital role in tennis.



figure 1: Wimbledon logo

1.2 Restatement of Problem

We need to analyze the attached data and answer the following questions.

1. Develop a predictive model to capture in-game scoring to assess player performance and visualize game conditions
2. Use the model to analyze the impact of "momentum" on the game.
3. Develop a new predictive model to predict the swings in the match, to find the main influencing factors, and to advise players on how to play with different players in a new game.
4. Test the developed model in other matches to predict the flow of volatility during the match and evaluate the model in terms of generalizability and other indicators.

5. Write a memo summarizing the results and advising the coach on the role of "momentum" and how to prepare the players for what to expect in the game.

1.3 Our Work

In order to better describe our work, we do so by drawing a flowchart, which is shown in Figure 2.

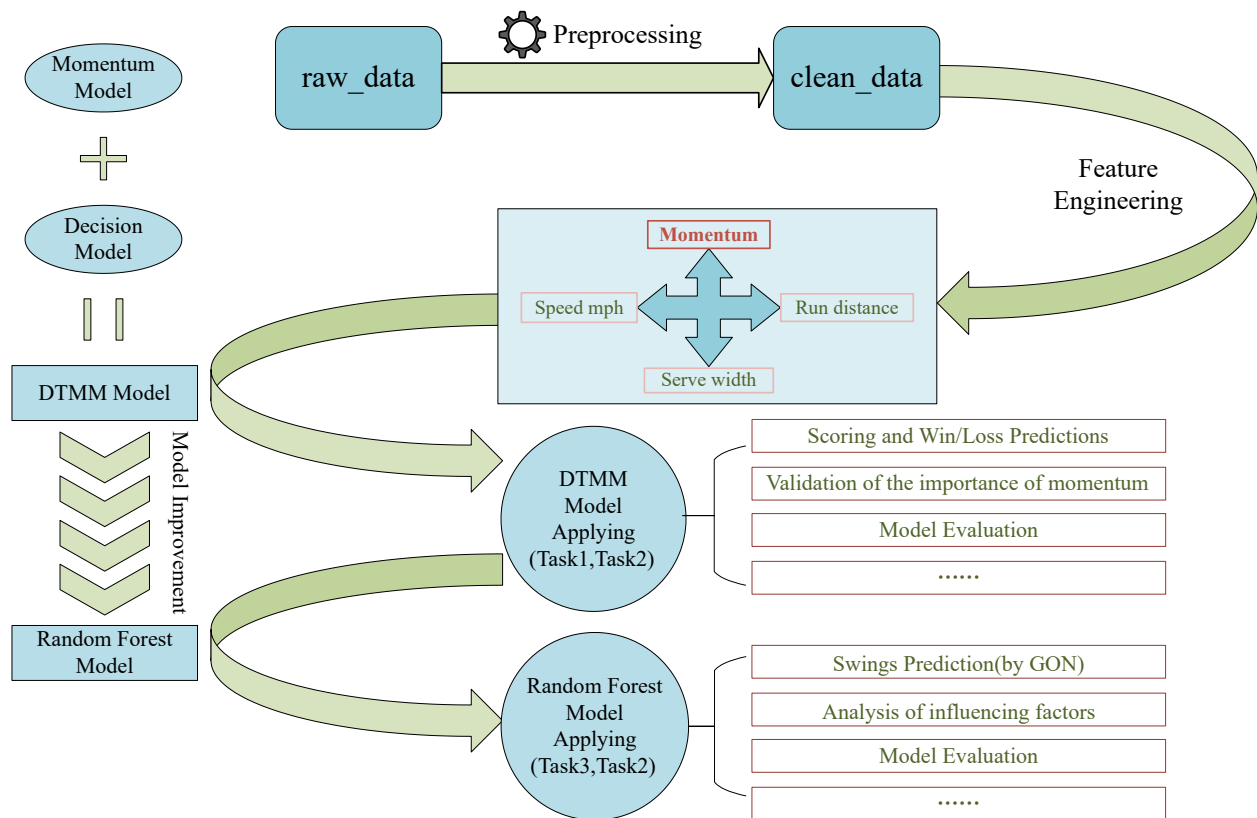


figure 2: Our Work

2 Assumptions and Notations

2.1 Assumptions and Justifications

In order to simplify our model, we make the following assumptions.

Assumption 1: Events that lead to incidental sudden changes in athletes' status were not considered.

Although an athlete's athletic state is related to many factors, to simplify the model, we assume that factors on the field alone determine the athlete's state and, therefore, do not consider the influence

of uncertainties such as spectators off the field and slippery fields.

Assumption 2: The objective level of the athlete does not change as the game progresses

During a match, an athlete may acquire new strokes and other skills or have a sudden "epiphany" that dramatically improves technique. However, professional players have been training for a long time, and their level of tennis has stabilized chiefly, and accounting for these contingencies complicates the model.

Assumption 3: Assuming that the advantage brought by two players serving alternately is the largest among the influencing factors affecting the momentum, its weight is recorded as the maximum value of the remaining factors.

On one hand, from the information given in the question and the tennis experience, whether to serve or not plays a significant role in winning or losing the match, so its influence on the momentum should also be significant; on the other hand, since the entropy value of whether to serve or not to serve is minimal in entropy weighting method, it is not unreasonable to use the entropy weighting method together with other factors.

2.2 Notations

table 1: The list of Notations

Symbol	Definition
M	Momentum for tennis players
$p1_{actual}$	The ratio of points won by Player 1 in a match to the total number of points won.
$p2_{actual}$	The ratio of points won by Player 2 in a match to the total number of points won.
$p1_{predict}$	The ratio of points won by player 1 in a match after model prediction to total points won data
$p2_{predict}$	The ratio of points won by player 2 in a match after model prediction to total points won
$V_{actual}^{\rightarrow}$	The 0-1 vector consisting of the code names of the actual winners of each point in a game
$V_{predict}^{\rightarrow}$	The 0-1 vector consisting of the code names of each point predicted the winner of a match.

3 Data Preprocessing

In order to deal with outliers and missing values in the data and to feature engineer the raw data into something that can be utilized, we do the following.

- We drop this feature since the data in the return_depth column has too many missing values.
- Since most of the variables in batting are continuously varying, such as ball speed and width. Therefore, we utilize the average of two adjacent data values to replace missing values.

- Different directions and distances from the baseline generally correspond to different scoring probabilities, so we convert different directions and distances from the baseline into numbers between 0 and 1 according to their scoring advantages, and the specific mapping relationships are shown in Table 2

table 2: Mapping of Serving Behavior to Scoring Chances

serve_width	W	0.3
	BW	0.25
	B	0.2
	BC	0.15
	C	0.1
serve_depth	CTL	0.8
	NCTL	0.2

4 Task 1 and Task2: Calculating Momentum and Modeling decision trees

In this section, we developed a DTMM model to characterize the effect of factors such as a tennis player's momentum on match scoring. This, in turn, answers the first two questions.

4.1 Momentum Calculation in Conjunction with The Entropy Weight Method

4.1.1 Momentum modeling

Considering that momentum is a subjective concept that is difficult to measure, we will select some factors affecting momentum and find out their importance through the entropy weighting method and normalize and normalize them, after which we will add them up and then get the empirical formula for measuring momentum.

In tennis, momentum is related to many players' psychological factors, and it is often the on-court game situation that leads to their psychological changes, so we chose six indicators as the parameters of our model, i.e., `points_advantage`, `break_points`, `unforced_errors`, `winner`, `net_pt_won`, `serve_advantage`. denoted by x_1 , x_2 , x_3 , x_4 , x_5 and x_6 .

Among them, `points_advantage`, `break_points`, `unforced_errors`, `winner`, `net_pt_won`, and `serve_advantage` are all obtained from the initial data calculation and processing, and their meanings are shown in the table below:

points_advantage: the difference between all the points won by the player.

break_points: the sum of points won on the opponent's serve.

unforced_errors: The sum of points lost due to unforced errors.

winner: the sum of points won by hitting an untouchable winning shot

net_pt_won: the sum of points won while at the net (the sum of points gained by scrubbing the net)

serve_advantage: the sum of points won on your serve

Since the parameters we chose are dimensionless normalized data and the data already fall in the interval $[0,1]$, no normalization, standardization, or normalization is required. Subsequently, we use the entropy weighting method to calculate and obtain the weights, denoted as a_1 , a_2 , a_3 , a_4 , a_5 and a_6 . The specific values are shown in Figure 3.

	points_advantage	break_points	unforced_errors	winner	net_pt_won	serve_advantage
p1	0.151043744	0.163629272	0.161450939	0.177812494	0.168251057	0.177812494
p2	0.1524303491	0.165761182	0.162859366	0.172198497	0.172678968	0.172198497

figure 3: Heat map of the weights of the parameters in the momentum calculation

The formula for calculating momentum is as follows:

$$M = a_1x_1 + a_2x_2 + a_3x_3 + a_4x_4 + a_5x_5 + a_6x_6 \quad (1)$$

4.1.2 Momentum visualization and analysis of results

Through Equation (1), we calculated the value of momentum over time, and because of the large sample size, we chose the data of the first 100 seconds of the match with the "match_id" of "2023-wimbledon-1701" and plotted the momentum curves of the two athletes in Fig.4 at the same time.

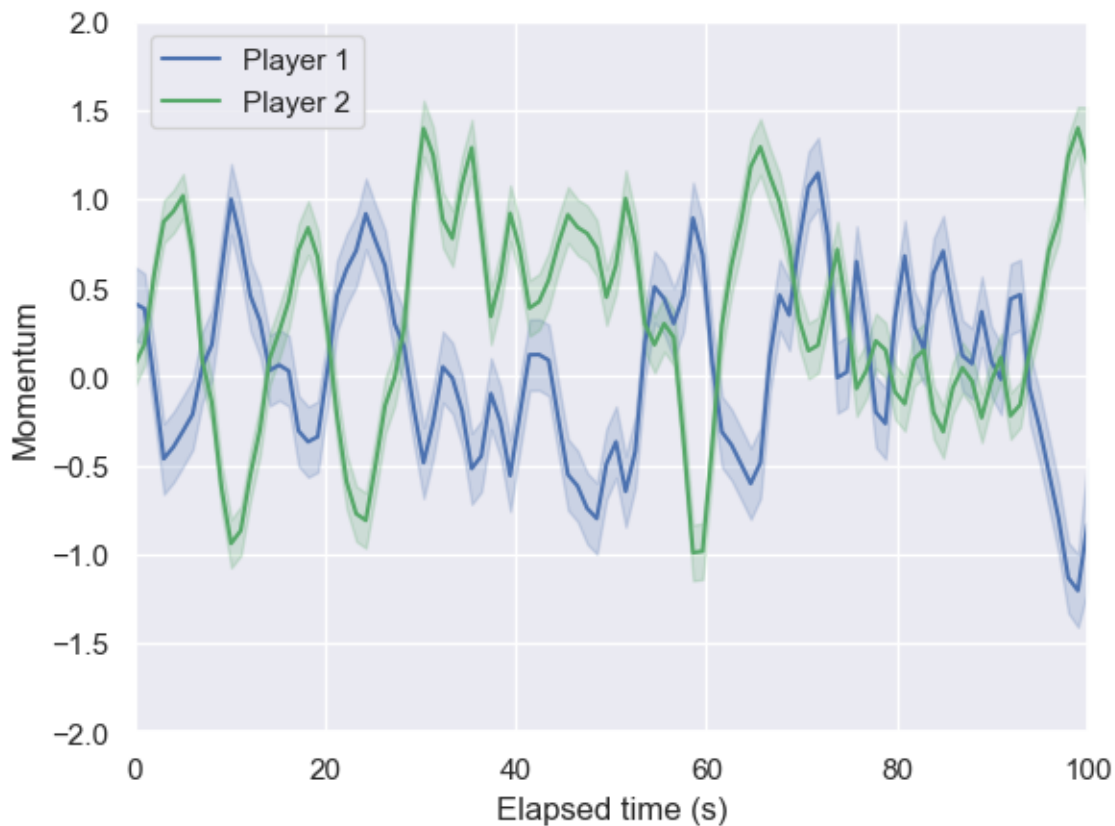


figure 4: Momentum over time curve

For this image, we get the following conclusions:

- For each movement, his momentum shows periodic fluctuations over time.
- For two athletes, their momentum is characterized by this and that. Not surprisingly, in competitive sports such as tennis, when one athlete has a high momentum, it often means that he is outperforming the other athlete in the game so that his opponent is relatively frustrated and as a result, he has less momentum.
- For each athlete, most of their discrete momentum values are concentrated near the two extremes, similar to the 0-1 distribution of match wins and losses, from which it can also be seen that the two have some correlation, and the detailed arguments will be given in the following tour analysis and the parameter importance analysis of the decision tree model.

4.1.3 Run Test

After obtaining the weights, as shown in Fig.3, to verify whether the change of momentum during the game is random and whether the runs of success under the influence of momentum also appear

randomly, we conducted a randomness test on these two data sets. First of all, we have to get the data to measure the runs of success; Here, we quantify the fluctuating changes in the number of player wins by finding the number of wins in a time window. From this, we obtained momentum data for players one and two and Fluctuating data on the number of wins. Subsequently, using SPSSPRO's built-in tour test model, we examined the data for a total of 29 games and presented the results in Table 3

table 3: p-value after run test for win/loss results for all matches

	p_1	p_2	p_1 <i>match_swing</i>	p_2 <i>match_swing</i>
2023-wimbledon-1301	0	0	0.073	0.952
2023-wimbledon-1302	0	0	0.406	0.832
2023-wimbledon-1303	0	0	0.249	0.077
2023-wimbledon-1304	0	0	0.208	0.335
2023-wimbledon-1305	0	0	0.653	0.309
2023-wimbledon-1306	0	0	0.002	0.435
2023-wimbledon-1307	0	0	0.699	0.022
2023-wimbledon-1308	0	0	0.017	0.059
2023-wimbledon-1309	0	0	0.724	0.086
2023-wimbledon-13012	0	0	0.783	0.35
2023-wimbledon-13013	0	0	0.397	0.404
2023-wimbledon-13014	0	0	0.346	0.751
2023-wimbledon-13015	0	0	0.946	0.041
2023-wimbledon-13016	0	0	0.8	0.176
2023-wimbledon-1401	0	0	0.435	0.287
2023-wimbledon-1402	0	0	0.071	0.297
2023-wimbledon-1403	0	0	0.125	0.671
2023-wimbledon-1404	0	0	0.35	0.062
2023-wimbledon-1405	0	0	0.01	0.916
2023-wimbledon-1406	0	0	0.01	0.916
2023-wimbledon-1407	0	0	0.003	0.065
2023-wimbledon-1408	0	0	0.012	0.002
2023-wimbledon-1501	0	0	0.011	0.894
2023-wimbledon-1502	0	0	0.064	0.064
2023-wimbledon-1503	0	0	0.136	0.012
2023-wimbledon-1504	0	0	0.22	0.938
2023-wimbledon-1601	0	0	0.14	0.086
2023-wimbledon-1602	0	0	0.586	0.635
2023-wimbledon-1701	0	0	0.143	0.495

The above are the p-values of the athletes' momentum and fluctuating data on the number of wins after the swim test for the twenty-nine races. In the swim test, we first assume that all processes are stochastic, so the original hypothesis is rejected at $p < 0.05$, which means that the process is non-stochastic and, conversely, that the process is stochastic.

From Table 3, we can see that on one hand, the test results of momentum data for both p_1 and p_2 are 0, which means that momentum is not randomly occurring data; on the other hand, the test results of momentum-difference data are all greater than 0.05, which means that it is randomly occurring data.

Since momentum tends to increase after a match victory and vice versa, it decreases. Therefore, fluctuating data on the number of wins can represent the trend of winning and losing the match. Moreover, fluctuating data on the number of wins is random, so runs of success are also random.

To summarize, momentum - i.e., "swings in play" - appears not random in a match, while runs of success are randomly.

4.2 Score Prediction Based on Decision Tree

In this section, we build the DTMM (Decision Tree Momentum Model) model to make predictions about the scoring of tennis matches. Since momentum plays a vital role in winning and losing tennis matches, we choose a series of indicators centered on momentum as the decision tree nodes to analyze and predict.

4.2.1 Description of The DTMM

For this question, we choose the decision tree model, on the one hand, because we want to predict the result of the scoring situation; it is the discrete dependent variable, at the same time for the number of goals scuffed net and other independent variables are also discrete, which continuity of the momentum factor can also be discretized by way of staging, so the independent variables are also mostly discrete, so we choose the decision tree model.

A decision tree is an essential machine learning model for classification and prediction, essentially a model that generalizes a set of classification rules from a training set and uses the training data set to estimate conditional probabilities. It utilizes a recursive algorithm as well as Equation (2) to compute the information gain, which determines the nodes' placement.

$$\begin{aligned}
 H &= -\sum_{i=1}^n p(x_i) \log_2 p(x_i) \\
 H(D) &= -\sum \frac{|c_k|}{|D|} \log_2 \frac{|c_k|}{|D|} \\
 H(Y|X) &= \sum_{i=1}^n p_i H(Y|X = x_i) \\
 g(D, A) &= H(D) - H(D|A)
 \end{aligned} \tag{2}$$

Notice: In the above four equations, the information expectation, empirical entropy, conditional

entropy, and information gain of the data are calculated respectively

We also use the id-3 pruning algorithm for model optimization to prevent the model’s overfitting. For better presentation, we only chose a 4-layer decision tree structure as an example, as shown in Fig.5.

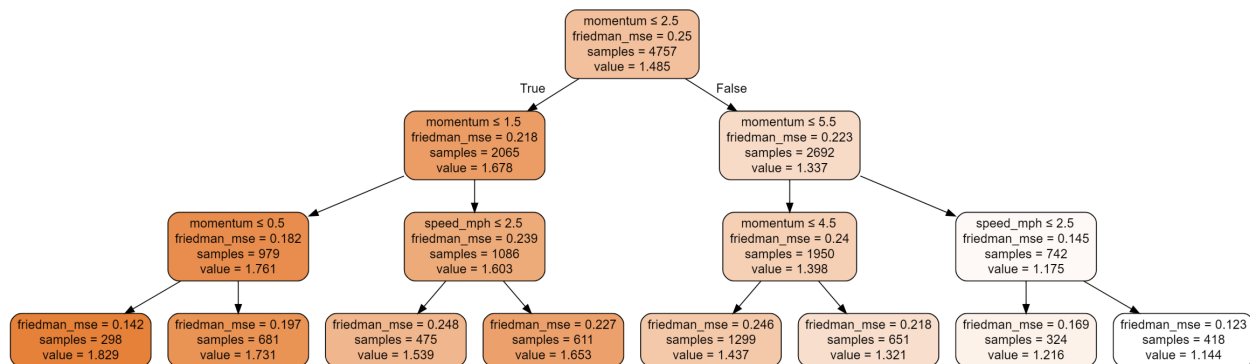


figure 5: Sample decision tree for $n=4$

4.2.2 Feature Selection

We classify the parameters into the following two main categories for node feature selection in decision trees.

1. Momentum features:

To better measure the relationship between the strengths and weaknesses of the two athletes, we chose the momentum difference between the two athletes as a feature.

In sports, momentum is a difficult-to-measure force that an athlete may believe he or she possesses during a game. In our model, it has been transformed into a form that can be discussed quantitatively through Equation(1).

Since the decision tree model requires the variables to be discrete, we made a histogram of the frequency distribution of the momentum difference, counted its distribution of the momentum difference, and, based on this, divided the momentum difference into eight data classes.

On the other hand, when plotting the histogram of the frequency distribution of the momentum difference, we accidentally found that it roughly follows the normal distribution, which will be the direction of our future exploration, and the results are shown in Figure6.

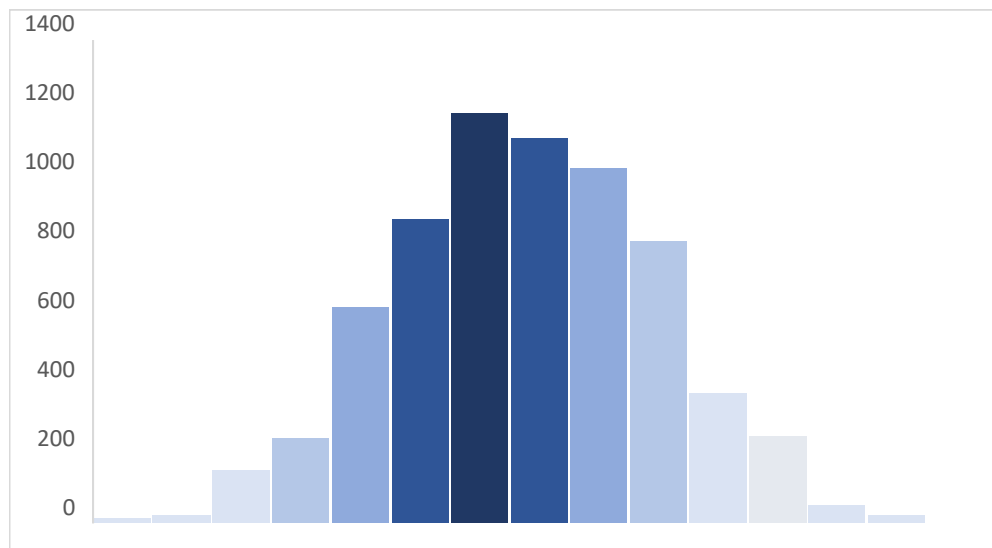


figure 6: Quantitative distribution of different values of momentum

2. Other features

Momentum tends to affect the state of the athlete's subjectivity more than anything else. To make the model more complete and improve its compatibility and generalizability, we therefore chose to analyze three other factors, all of which are objective. Namely: run distance, serve width, speed mph

We organize and summarize these factors in Figure7

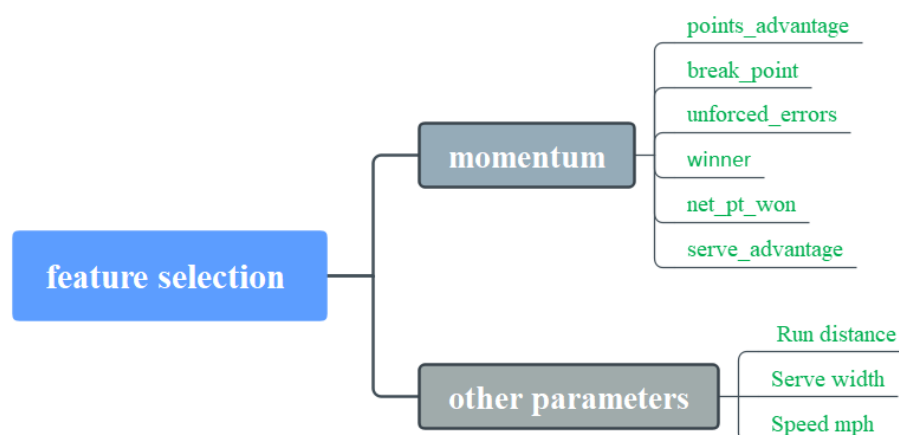


figure 7: Mind map of influencing factors

4.2.3 Model Construction And Data Analysis

1. Model Construction

Since the speed_mph attributes of the races with match_id "2023-Wimbledon-1310" and "2023-Wimbledon-1311" are missing in the original data, we do not use these two races in our prediction model for training and prediction. Therefore, the data of these two races are not used in our prediction model for training and prediction.

Subsequently, we conducted the following three experiments on the cleaned data.

- In the first experiment, we chose the date of the last eight games as the prediction set and randomly divided the remaining data into the train set and validation set in the ratio of 7:3, trained our decision tree model, and performed post-pruning operations, and finally obtained the decision tree model DTMM with momentum as the core parameter.
- In the second experiment, we take the data of the first six matches as the prediction set and do the same analysis as the first experiment.
- In the third experiment, to verify our model's robustness, we deliberately chose "2023-Wimbledon-1701" as the prediction set, which is perceptually more unexpected.

2. Data Analysis

We predicted the results of point_vector using the DTMM model we just obtained; based on the prediction results, we calculated and obtained the $P1_{actual}$, $P2_{actual}$; $P1_{theory}$, $P2_{theory}$ for each of the three trials, and the results of player2 are displayed in Fig.8 and Fig.9

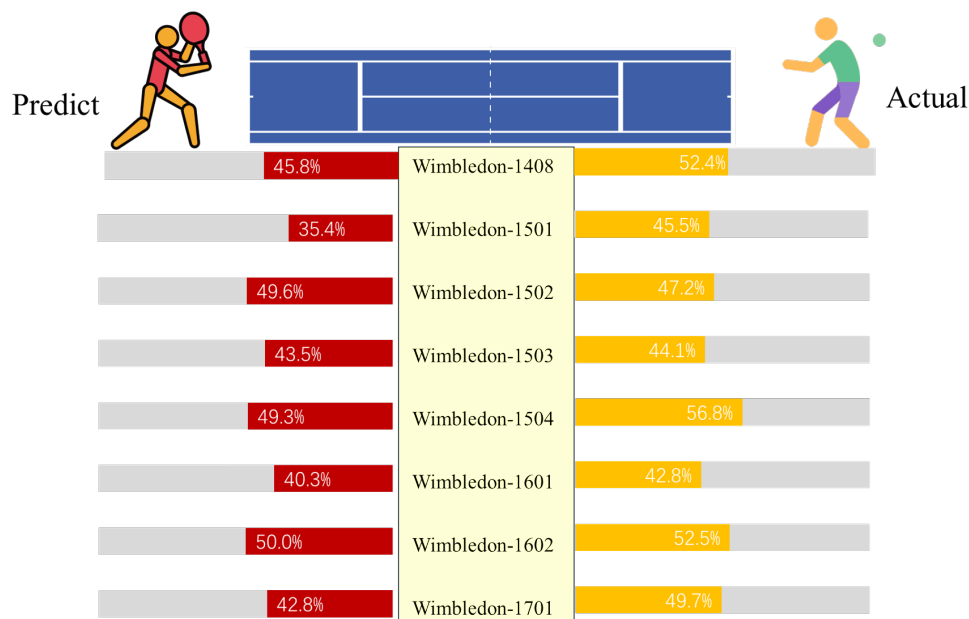


figure 8: Player 2 in the second eight games' win/loss result prediction figure.

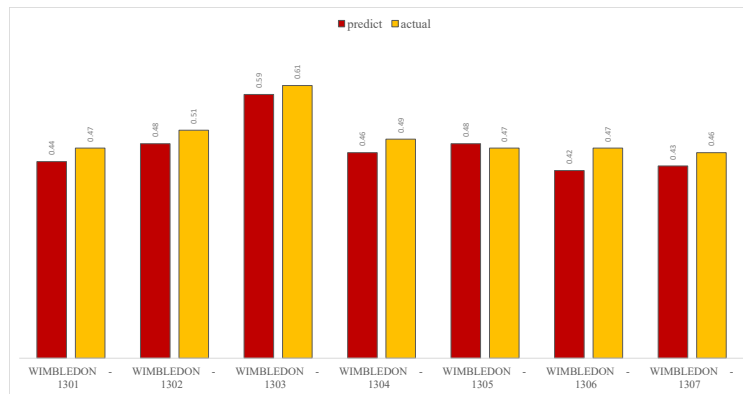


figure 9: Player 2 in the first seven games' win/loss result prediction figure.

In the third experiment, we predicted the last match and found that the match rate was 80%.

The above is the prediction in terms of a whole game; then we further refined the model by narrowing the prediction interval to 20 points, analyzed the game 'match-id '2023-wimbledon-1701' in detail and trained a decision tree, and subsequently plotted the predicted and actual values of the number of games won by player2 within 10points in Figure 10. We then performed a Pearson's correlation coefficient of 0.9192, which indicates that our model is also accurate in predicting over a small time horizon.

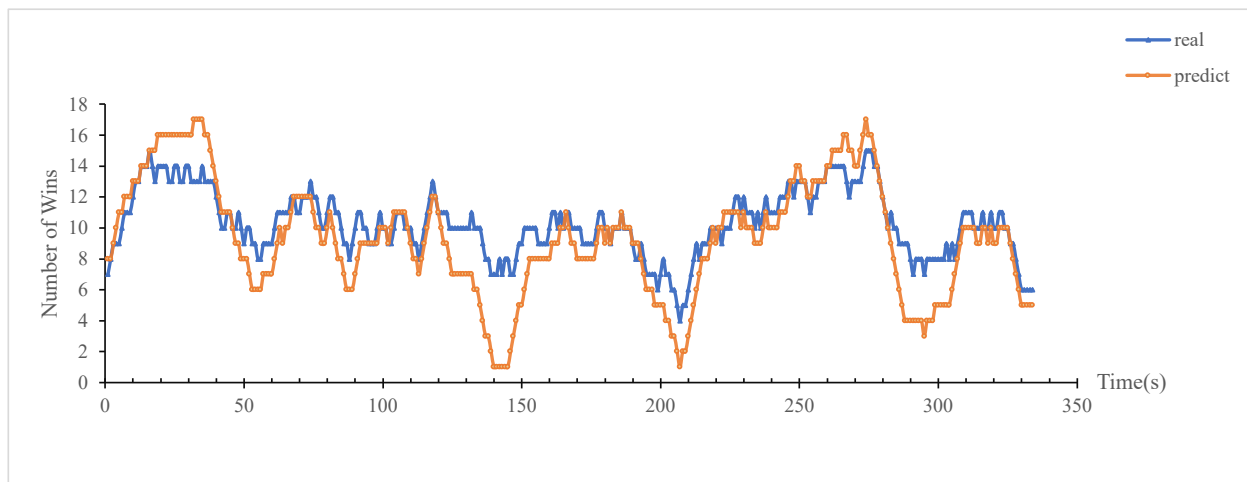


figure 10: Plot of change in model predictions vs. actual game wins for time window of 20

From the above results, it can be seen that our prediction of the winning percentage is very close to the actual value, with the percentage error ranging from 1% to 14%, which indicates that our model works well in predicting the winning and losing percentage of a game.

On the other hand, in order to explore the direct effect of momentum on the win/loss of the game, in our third experiment, we converted the win/loss vector $V_{predict}^{\rightarrow}$ and the actual win/loss vector $V_{actual}^{\rightarrow}$ for each point obtained from the player game prediction to unit vectors. We performed the cosine similarity test using Eqn.(3) and obtained a similarity of 95.898%, which indicates that the reutilization of our DTMM model predicts the match results quite accurately.

$$cos = \frac{V_{actual}^{\rightarrow} \cdot V_{predict}^{\rightarrow}}{\|V_{actual}^{\rightarrow}\| \cdot \|V_{predict}^{\rightarrow}\|} \quad (3)$$

Based on the above data, we found an interesting phenomenon. We briefly analyzed the correlation of momentum as well as momentum_diff. Applying the control variable method, when momentum is not used as an influencing factor in the decision tree, the predicted data match_rate for 1701 matches is only 59%, showing a significant decline. Relying on the data on the number of runs scored by player 2 per twenty overs, again controlling for momentum, the Pearson correlation coefficient drops dramatically to 5.9%. This suggests that "swings in play" correlate with "runs of success."

4.2.4 Error Analysis

In this study, the accuracy of each model was verified using the cross-validation method, and the root mean square error (RMSE) and mean absolute error (MAE) generated from the cross-validation results were used as the evaluation indexes for the accuracy of the estimated and verified models. Among them, the R2 metric is not used because it does not apply to evaluating discrete variables and machine learning models. Among the two selected metrics, the smaller the RMSE and MAE, the higher the model's prediction accuracy. We examined the prediction results for all the matches and displayed the results for two of them in Table4

table 4: Cross-testing of predicted win/loss results for matches 1301 and 1701

	MAE	RMSE
2023-wimbledon-1301	0.243	0.493
2023-wimbledon-1701	0.201	0.448

5 Task 3:Model Improvement and Feature Exploration

In this task section, we partitioned all the available match data and extracted all the match data of Novak Djokovic, which was summarized into a training set. After that, we predicted the volatility of two matches, 1316 and 1701, and analyzed the importance of each influencing factor on the volatility characteristics. Based on the above data, we obtained match recommendations for Novak Djokovic.

5.1 Match swings Analysis

5.1.1 Match Swings Definition

The swings of the game are an objective quantity, and the change in the game's score should measure it. Moreover, if we directly choose the win or loss of each point to measure, it may be disturbed by the abnormal play of a particular ball; in order to exclude this interference and improve the robustness of the model, we choose to sum the number of wins of 20 consecutive matches to get the wins of both sides of the match and the sequence of wins and the sequence of $S_{player1}, S_{player2}$ ($0 \leq S_{playeri} \leq 20, i=1,2$). One of the players is analyzed, and his GON (good or not) sequence is calculated using Equation (4).

$$GON = \begin{cases} 1, & S_{playeri} > 10 \\ 0, & S_{playeri} = 10 \\ -1, & S_{playeri} < 10 \end{cases} \quad (4)$$

5.1.2 DTMM improvement

1. Random Forest When we used the DTMM model for the prediction of the GON value, we found that the accuracy of the model decreased compared to task 1 for predicting the probability of winning a match, which may be because the dataset for predicting the probability of winning a match is relatively large, which is an entire match, and the basic unit for predicting the GON is 20 pts. We plan to optimize it by using the Random Forest algorithm.

For Djokovic's five matches, we choose four of them as the training set, and each time, there is put back from the selection of three matches as a sub-training set for model training, i.e., a total of four estimators were constructed. Based on the Bagging algorithm, a random forest model was obtained. The model was used to predict two matches, 1316 and 1701.

2. We show the results of the two-match predictions in Figure 11 and Figure 12, which shows the predicted and actual values of Djokovic's GON in the two matches, 1316 and 1701.

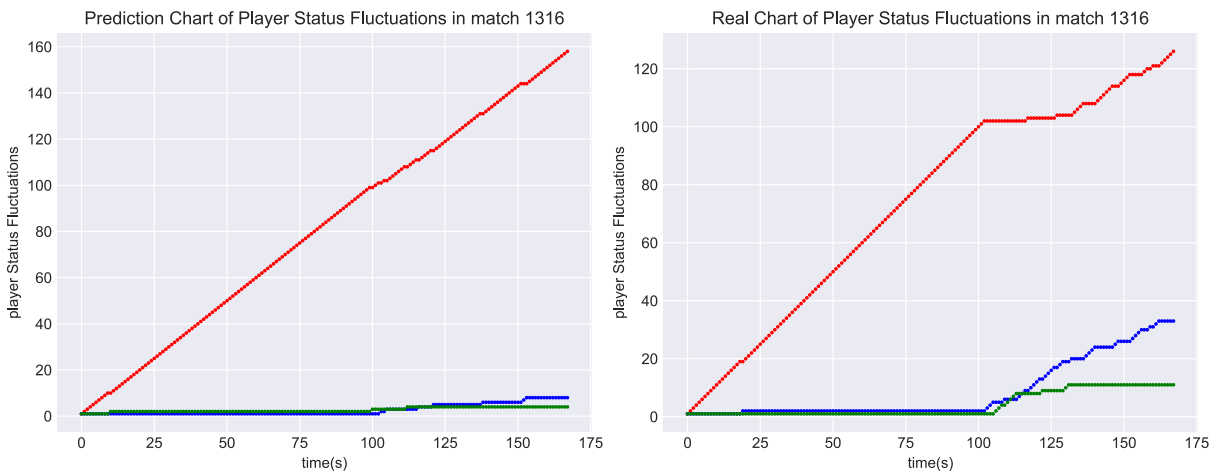


figure 11: Scatterplot of change in predicted vs. actual GON values for fields 1316

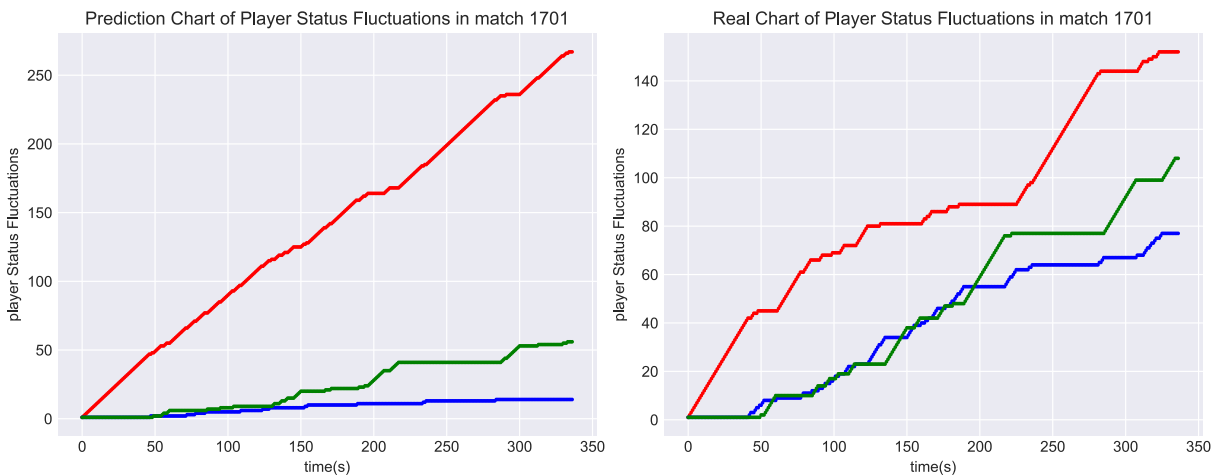


figure 12: Scatterplot of change in predicted vs. actual GON values for fields 1701

5.2 Analysis of influencing factors

In order to explore the importance of the factors influencing match fluctuations, we used Python's built-in shap library to find the shap values of the DTMM model we constructed and obtained the characteristic contribution values of each factor, forming the Figure13 .

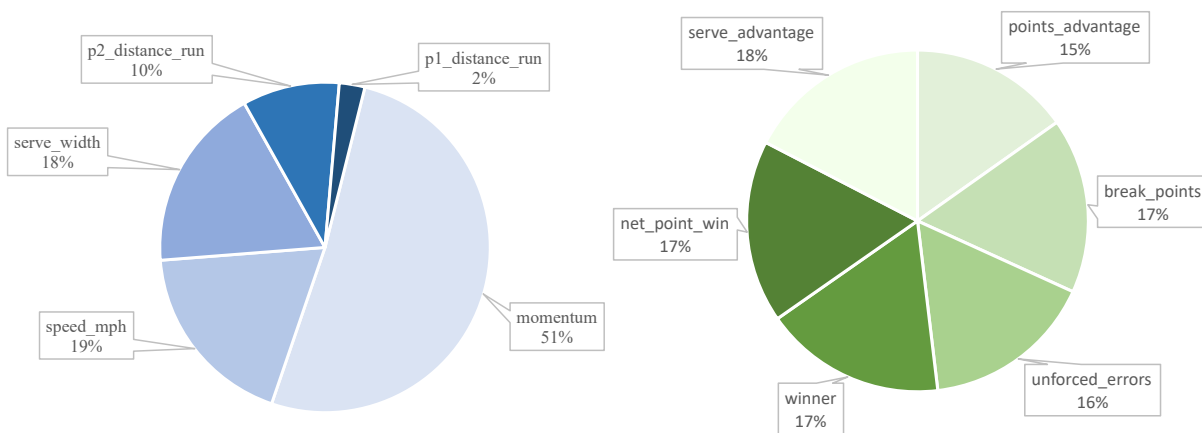


figure 13: Importance map of factors influencing match swings.

We then conducted a Pearson test for each factor with the following results:

- P2_points_advantage

Djokovic's scoring advantage over his opponents within the time window, which has a Pearson correlation coefficient of 0.3928 with momentum, close to a moderate linear correlation, which implies that the larger Djokovic's scoring advantage is, the better he performs on the field of play, which can be attributed to Djokovic's desire to win the game and positive psychological cues.

- P2_break_point_won

The sum of Djokovic's points on his opponent's serve within the time window, with a Pearson's correlation coefficient of 0.3947, which is close to a moderate linear correlation, basically implying that winning at a disadvantage is more motivating for Djokovic to achieve better performance.

- p2_unforced_errors

the sum of Djokovic's points lost due to unforced errors within the time window; its Pearson correlation coefficient with momentum is -0.2482, close to a weak linear correlation, which means that the more errors Djokovic makes, the more negative his performance is, which conforms to the straightforward perception of the game fluctuations. Perception of match fluctuations.

- The rest of the factors were tested and found to be not linearly correlated, which means that changing these factors within the time window does not significantly impact the momentum of the match and the players' performance. Then, based on the trend of tennis match development, the matches can be categorized into four scenarios; for Djokovic, the matches show an advantage, the matches show a disadvantage, the matches move from a disadvantage to an advantage, and the matches move from an advantage to a disadvantage.

5.3 Advice for Players

Above, we have analyzed five matches of Novak Djokovic only and given the following advice to tennis players based on his performance:

1. Sharpening serve and receive skills In tennis, the serving player has a tremendous technical advantage, and it often happens that the player scores points because of his serve or by breaking serve. Therefore, it is of great importance to cultivate excellent serving and receiving skills. In daily training, players should train the ability to control the landing point of the serve and improve the ability to control the speed of the serve to get a more significant advantage. In addition, the players should continuously strengthen their serve-receiving skills. Improve the players' judgment of the ball landing point and speed to improve the level of serve-receive technology.
2. Reduce unforced errors Players' mistakes on the court are affected by multiple factors, one of which is unforced errors, which refers to the mistakes made by players when they are not pressured or pressured by their opponents. These errors are usually caused by technical inaccuracies, errors in judgment, or lack of concentration rather than by an opponent hitting a quality return. Therefore, players should train in basic tennis skills to minimize low-level errors such as going out of bounds, hanging on the net, and going under the net.

- Strengthen the psychological construction From the above data analysis, P2_points_advantage also has a more significant impact on players' game performance. Often, players can maintain a good mindset when the score is ahead, but when the score is behind and the game is in trouble, the players' momentum drops significantly. Therefore, the psychological construction of players should be strengthened in daily training. Cultivate the ability of players to maintain their mentality when they are behind in the score and improve the stability of their gameplay.

6 Task 4:Model Migration

We downloaded all the data of the 2023 U.S. Open tennis tournament, divided the dataset into training and validation sets in the ratio of 7:3, and predicted the fluctuations of the tournament using the features in the original DTMM, of which 1,000 scoring intervals were selected, and plotted Fig.14 to reflect the fluctuation prediction of the tournament.

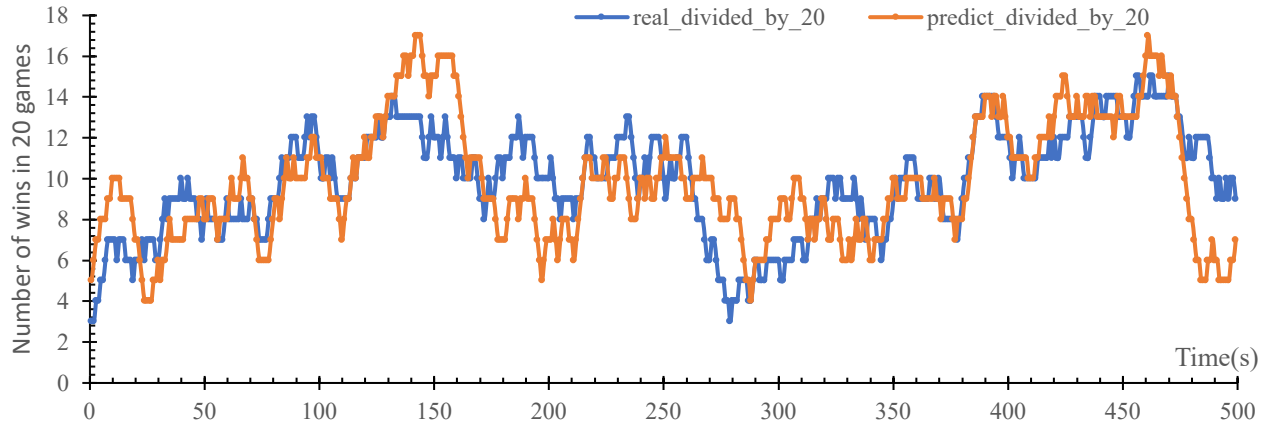


figure 14: 2023 U.S. Open Fluctuation Schematic

The Pearson correlation coefficient between the sequence of predicted values and the sequence of valid values is 0.676026056280571, which is a strong correlation, and it can be proved that our model can get better results in predicting the match fluctuation of the U.S. Open tennis tournament as well.

Subsequently, we calculated the variance of the true value series and the predicted value series, The results are shown in Table 5.

table 5: Variance of true values predicted values

V_{real}	$V_{predict}$
6.338	9.557

This shows that our model predicts a more excellent range of match fluctuations, which may be because we did not take into account the differences between the U.S. Open and Wimbledon, such as the difference in venues, the difference in the date of the match, and the weather, etc., which would increase the inaccuracy of the model prediction.

We also calculated the SHAP value of this decision tree to determine the contribution of each feature to this decision tree, and the results are shown in Figure 15.

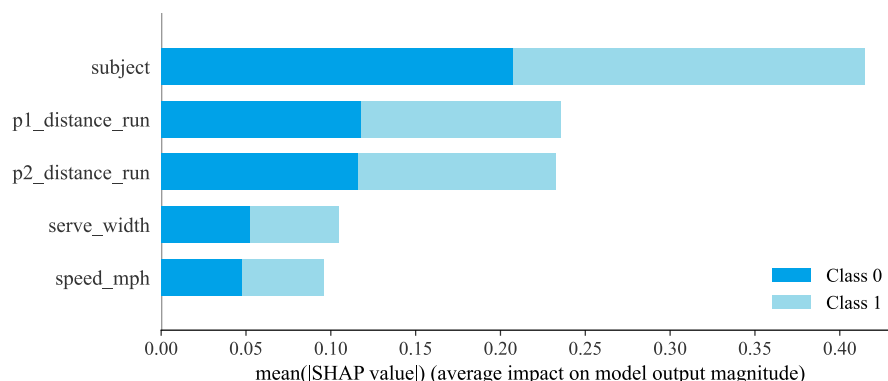


figure 15: Contribution of individual features to the model

In addition, if we want to construct a more accurate model to predict the fluctuation of the game, we may also need to consider the weather factors, such as temperature, air humidity, etc. Changes in the weather or temperature will affect the physical functions of the athletes, which will be the reason for the fluctuation of the game.

7 Sensitivity Analysis

In constructing the DTMM model, we perform a sensitivity analysis to explore the parameters' and complexity' effects on the model's predictive ability. Controlling the variables, the model prediction is performed only by changing the combination of the parameter cases and the tree body. In this way, we can use it to test the robustness of the model.

First, we explore the sensitivity of the model to the parameters. We reduced each of the four influencing factors separately, passed the remaining three parameters into our model, and randomly selected a game to predict to get the probability of its prediction being accurate and compared it with the probability of the model predicting accurately when four parameters were passed in.

Second, we explore the sensitivity of the model to model complexity. The initial value of the tree body was set to 50, and five layers of the tree body were added sequentially, for a total of five times, to obtain the probability of accurate model prediction for each time.

We visualize the results of the above two in Figure16.

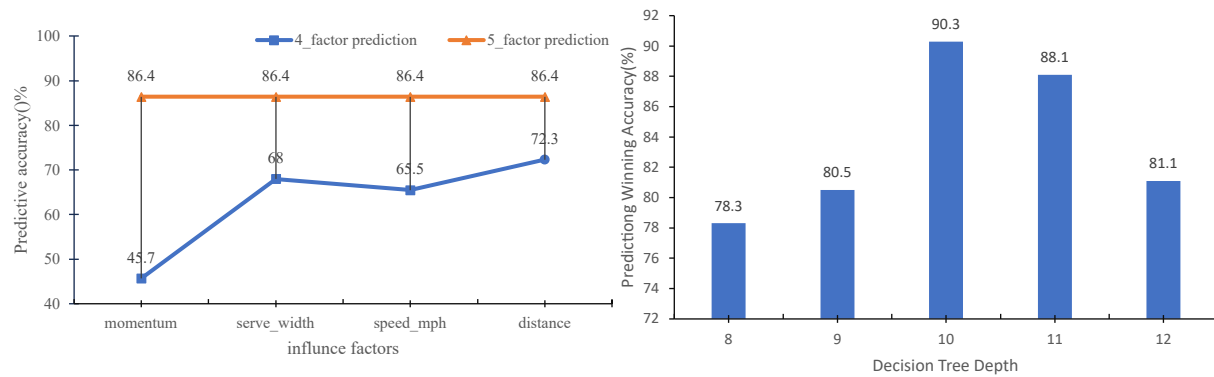


figure 16: Model predicted probability of success versus parameters (left panel) and tree body (right panel)

Among them, for the parameters, it is found that when the momentum factor is removed, the accuracy of the model decreases substantially. At the same time, there is a decrease when the other factors are removed, but they are all close to each other, and the decrease is not significant, which shows that the momentum is decisive for the accuracy of the prediction of the match wins and losses.

For the tree depth, we take twenty as the base tree body and calculate the prediction success probability when the tree depth is 8, 9, 10, 11, and 12 with it as the center and find that the accuracy rate increases and then decreases with the tree depth. The accuracy rate is the largest when the tree depth is 10, so we chose the tree depth of 10 when we built the model.

8 Model Evaluation

8.1 Strength

- The perceptual concept of momentum is quantified using a mathematical formula, and the entropy weighting method is used to give the weight of each parameter so that the formula can better describe momentum.
- The DTMM model predicts fluctuations in the number of points scored by an athlete in a given game, which is more realistic than traditional models.
- We carried out model nesting in constructing the DTMM model. We passed the constructed momentum model into the decision tree model as a node, which can give full play to the roles of different models and improve the scientific robustness of the models.

8.2 Weakness

- Due to time constraints, our consideration of the factors affecting the momentum formula when it was established may have needed to be more comprehensive, and more indicators can be collected to refine the formula in the future.

- The amount of training data when we perform model training is not very large, which may result in a model that could be more accurate.

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MEMORANDUM

To: tennis coach

From: Team 2417408

Date: Monday, February 5, 2024

After four days of work, we built the momentum calculation, DTMM, and random forest models. We integrated them organically and successfully analyzed the role of momentum in the game and its impact on the game.

Here, we present our research results as a paper and give suggestions.

In calculating the momentum, we applied the formula and got the quantitative analysis results of the momentum. Then, we selected the critical data in the game and built a machine-learning model with momentum to predict the game's flow. Momentum has a non-negligible impact on the result of the game. Through the randomness test, we found that the momentum in the player's game does not randomly appear; in addition, the momentum contributes more than half of the importance in the feature importance analysis.

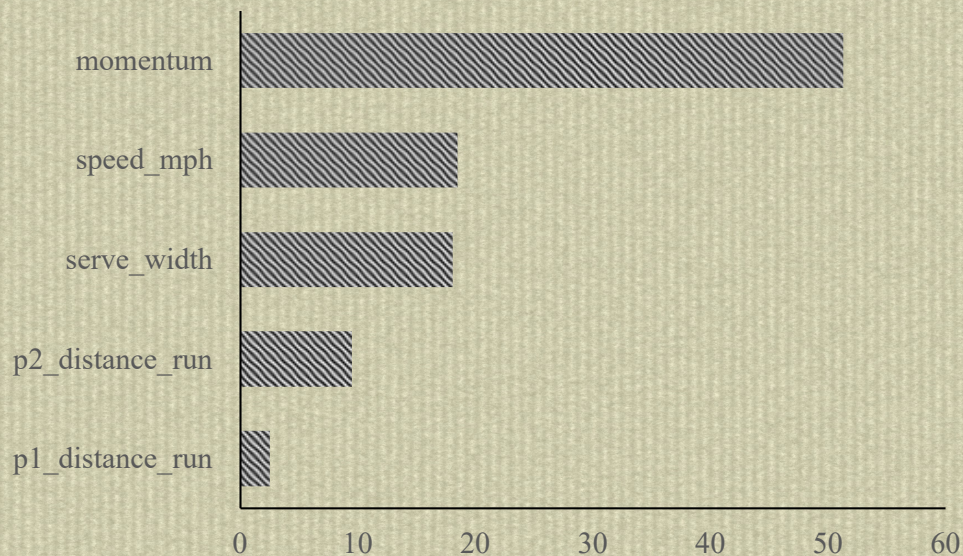


figure 17: Factors affecting the volatility of the game

This shows that the players' momentum can often be the decisive factor in a game. Mobilizing players and keeping them in good momentum is crucial to win or lose the game. Considering

the impact of momentum and other factors, we take Djokovic as an example and give the following suggestions:

1. When the game's momentum is favorable, i.e., when `points_advantage` is positive, you can pursue the victory, i.e., expand `points_advantage` to win the game as soon as possible, reduce your physical exertion, and gain more energy for the subsequent games.
2. When the momentum of the game is at a disadvantage, Djokovic should try to win his opponent's serve as much as possible, which statistically has a positive impact on his game performance, and should reduce his unforced errors, to avoid boosting his opponent's confidence and to have a positive impact on himself.
3. when the momentum of the game from disadvantage to advantage, Djokovic's `points_advantage` is still negative, it is easy to know that the rest of the factors that make the momentum reversal, then Djokovic should suggest that they do not focus on scoring, but rather focus on maintaining the current state of affairs
4. When the momentum of the game changes from advantage to disadvantage, Djokovic's `points_advantage` is positive, but other factors combine to make the momentum change from advantage to disadvantage, which means that his state is not good. This state is mainly caused by factors such as `break_point_won`, `unforced_errors`, etc., so Djokovic's strategy, in this case, is the same as when the game's momentum is harmful; there is no need to put too much pressure on him.